Exploring DoRA: Weight-Decomposed Low-Rank Adaptation Across Modalities

Richie Xue • Shaurya Sen • Kyle Du

Cornell University · Dept. of Computer Science · CS 4782: Introduction to Deep Learning · Spring 2025

Introduction

Fine-tuning billion-parameter models is expensive for most teams.

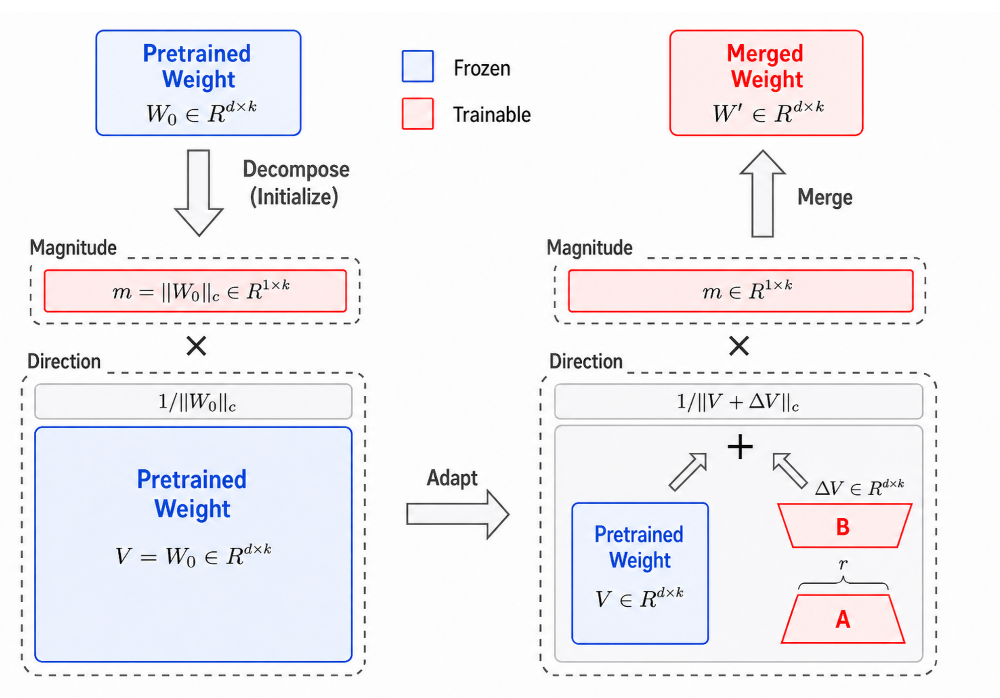
LoRA (Hu et al., 2022) helps by freezing the base weights W_0 and learning a low-rank update BA, but this update changes a layer's *magnitude* (how strong) and *direction* (what feature) **together**.

DoRA (Liu et al., 2024) decouples them: it learns weight magnitude separately while using LoRA to update the direction.

Goal: Recreate DoRA NLP paper results + extend to non-NLP modalities.

Methodology: 4 models, 6 benchmarks for LoRA/DoRA direct comparison.

DoRA Architecture



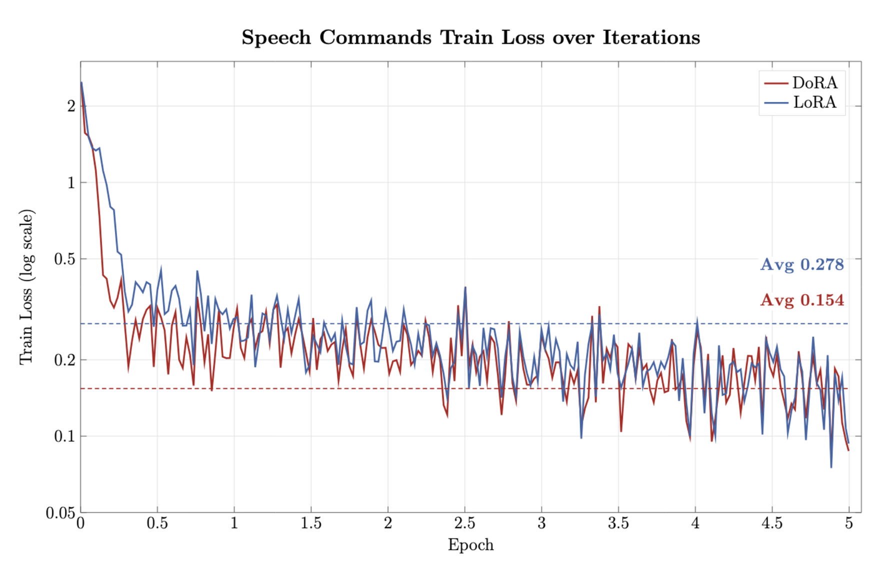
DoRA splits each weight W_0 into magnitude m and direction $V/\|V\|$, applying a LoRA update BA to the direction while training magnitude separately.

Audio Results

Wav2Vec2-base (67M params) fine-tuned on Google Speech Commands v0.02 - 84.8k utterances, **12 keyword classes**.

Method	Val Acc	Test Acc	Trainable	Time
DoRA ($r = 8$)	**98.6%**	**89.7%**	826.6k	**183.6 min**
LoRA ($r = 8$)	98.5%	89.0%	789.8k	190.8 min

+0.7% test accuracy. *The magnitude scalar's small overhead is offset by faster directional gradient convergence on the audio encoder.*



-44.6% avg tr-loss. *DoRA faster optimization and smoother convergence.*

NLP Results

Cross-Task at $r = 8$: (RoBERTa-base accuracy / F1+acc / accuracy)

Task	DoRA	LoRA	FFT	Δ
SST-2 (67k, sentiment)	93.2%	93.3%	93.2%	−0.1
MRPC (3.7k, paraphrase)	87.8%	85.9%	74.8%	+1.9
RTE (2.5k, inference)	**70.0%**	66.1%	54.9%	**+3.9**
SST-2 *large*	96.4%	95.5%	95.4%	+0.9
MRPC *large*	89.3%	89.5%	80.6%	−0.2
RTE *large*	76.9%	78.7%	68.6%	−1.8
OpenLLaMA-3B	81.0%†	—	—	—

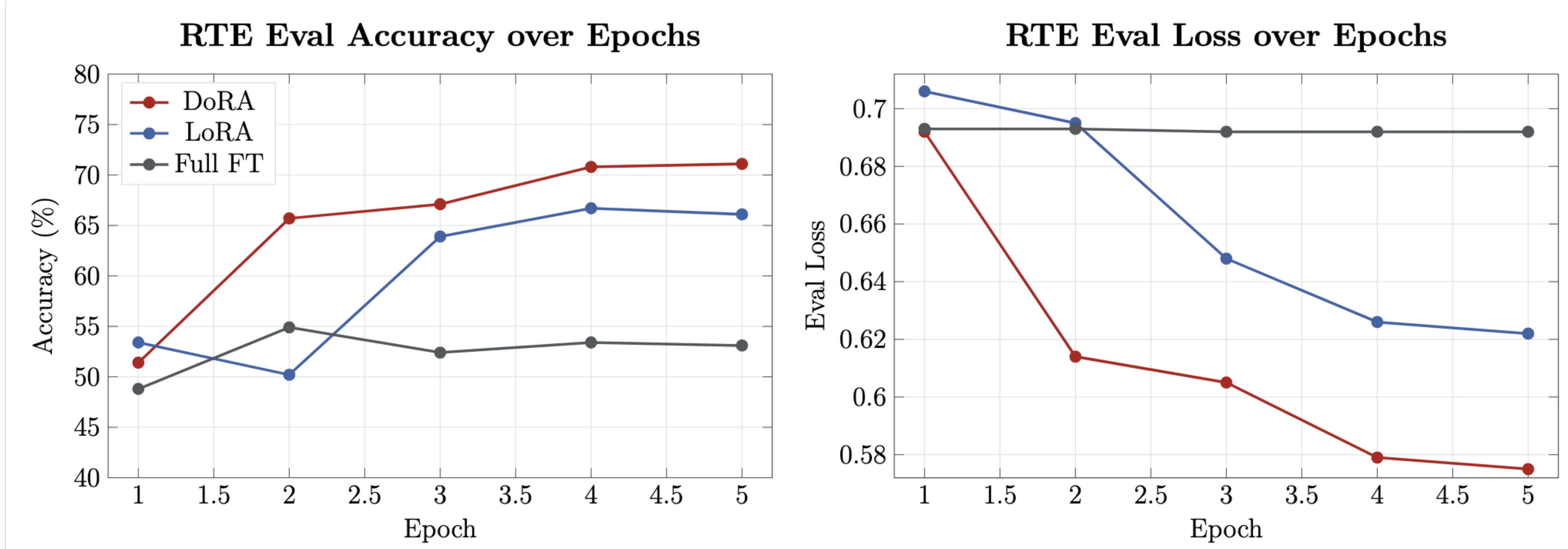
DoRA's advantage is largest on **low-data tasks** *(RTE +3.9 pp) where magnitude decoupling reduces directional gradient interference. Matches original paper results.*

Benchmark Task Examples: (Text → Label)

SST-2 *"...crowd-pleasing as a documentary can get."* → **Positive**

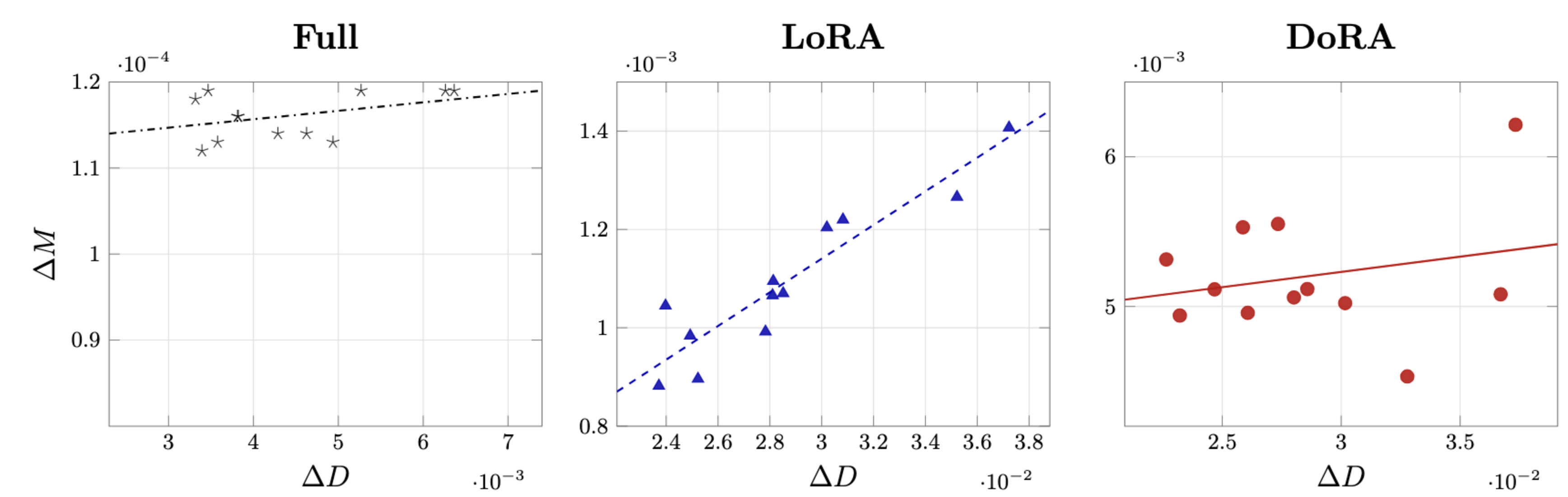
MRPC S1: *"Corixa rose 54c to $7.74."* S2: *"...rose 54c, or 8%, to $7.74."* → **Equiv.**

RTE P: *"Leaders signed the Egypt–Israel treaty in 1979."* H: *"War started in 1979."* → **Not Entailed**



DoRA converges faster and to a lower eval loss than LoRA on RTE, while full fine-tuning stalls near the random-guess baseline.

Magnitude-Direction Updates: (Full vs. LoRA vs. DoRA)



corr: Full = 0.40, LoRA = 0.93, DoRA = 0.24. *DoRA shows significantly lower slope and correlation than LoRA, closer to FFT, showing the effect of de-coupling* **M** *and* **D**.

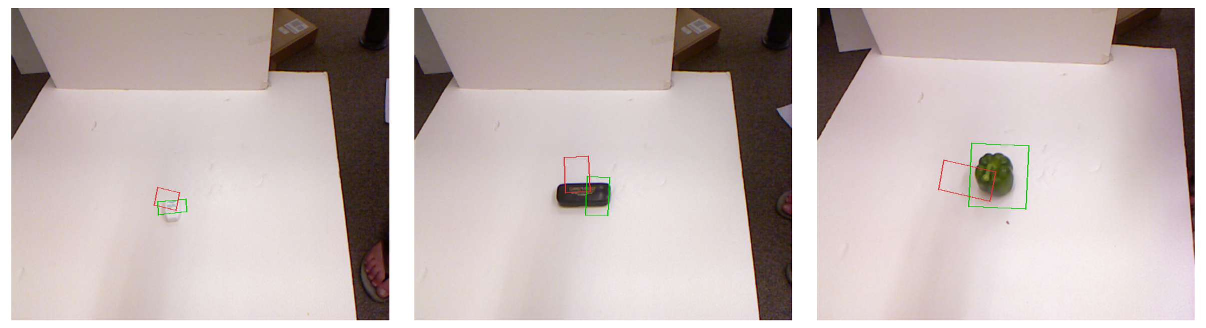
Vision & Robot Results

Object Grasping: Predict a 2-D bounding box around the graspable object in an RGB tabletop image. Frozen ViT-B/16 and SigLIP-B/16 backbones, adapted at r=8, scored by IoU.

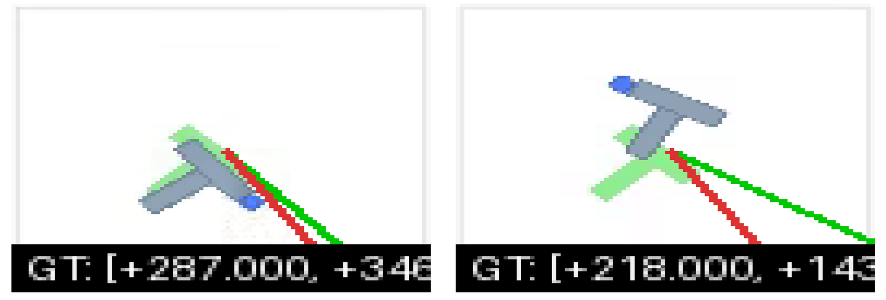
Backbone	DoRA	LoRA
ViT-B/16	11.9% IoU	11.9% IoU
SigLIP-B/16	**22.0%** IoU	21.5% IoU

SigLIP's richer vision–language pretraining benefits more from magnitude decoupling. ViT features show minimal difference.

Qualitative: SigLIP + DoRA: Green = ground truth Red = predicted grasp



Push-T VLA Task: A robot slides a T-shaped block to a target pose from an overhead camera plus language instruction. DoRA adapts a 126-layer OpenVLA model; action MSE drops 63.6 → **27.9** over 3 epochs.



DoRA adapter driving Push-T rollouts

Conclusions

- **Low-data NLP:** DoRA gains most when data is scarce — RTE +3.9 pp, MRPC +3.0 pp at r=16.
- **Audio:** +0.7% test acc. and 10% faster training on Wav2Vec2 — directional convergence is faster.
- **Vision:** SigLIP amplifies DoRA's benefit (+0.5 pp IoU); ViT-base does not, meaning base models matter.

Future Work

- **QLoRA + DoRA:** 4-bit quantized fine-tuning to reach 7B+ models on consumer hardware.
- **Diffusion models:** apply DoRA to action-distribution heads and image-generation U-Nets.
- **SVD-based decomposition:** explore EDoRA [7] — uses SVD to decouple singular directions, potentially reducing the parameter budget beyond DoRA.

[1] Liu et al. (2024). DoRA. *ICML*. [2] Hu et al. (2022). LoRA. *ICLR*. [3] Wang et al. (2018). GLUE. *EMNLP*. [4] Warden (2018). Speech Commands. *arXiv*. [5] Kim et al. (2024). OpenVLA. *arXiv*. [6] Jiang et al. (2023). SmolVLM. *HuggingFace*. [7] Nasiri & Garraghan (2025). EDoRA: Efficient Weight-Decomposed Low-Rank Adaptation via SVD. *arXiv*:2501.12067.